Running Case (HybridQA): *"What was the original name of the city with an AFL club formed in 1989?"*
Target Ground Truth: "York" • **Modality Fusion:** Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)

Stage 1 (Sec. 3.1): Multi-Source Ingestion

Table: AFL_Ontario_0

Club	Formed	City
Tor. Eagles	1989	Toronto
Ham. Tigers	1991	Hamilton

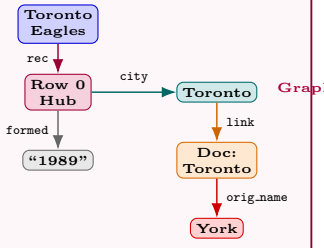
→ Matches constraint: **1989**

Wikipedia Bridge Passage:
Toronto → Doc:Toronto
*"...capital of Ontario. Established in 1793 as town of **York**..."*

Retriever Output
- Candidate table rows
- Linked Wikipedia passages
- Complete context retained

Evidence

Stage 2 (Sec. 3.2): Multi-graph Construction



Star Record Hubs
- Hub binds every cell to its row
- Prevents cross-row conflation
- Rule-based text links (no LLM)

Guided Reasoning Path
- Best path: Eagles → Row 0 → Toronto → York
- Guides model attention (no filtering)
- Compact serialized prompt

Graph + Path

Stage 3 (Sec. 3.3): Two-Tier Controlled Engine

Tier 1: Primary Reader [LLM Call 1]
- **Prompt Packaging**: Compressed graph (deduplicated passages + row blocks) + guided reading path
- **Controlled Decision**:
  - Clear evidence → **Fast Exit Answer** (1 call total)
  - Ambiguity or missing facts → **Diagnostic Handoff**

Fast Exit

if collision / ambiguous

Diagnostic Handoff: {"focus": ["Row 0", "Row 1"], "check": "original name"}

Targeted Bridge Retrieval [0 LLM Calls]: Fetches missing passages for disputed rows; builds focused capsule

Focused Capsule

Tier 2: Targeted Reader [LLM Call 2]
- **Prompt Packaging**: Transmits *only missing & disputed facts* (discards all unrelated rows)
- Resolves candidate ambiguity via verified facts ($\leq$ 2 calls total)

Tier 2 Verified

Stage 4 (Sec. 3.4): Grounded Provenance

Verified Answer Span
"York"
- Exact context-grounded extraction
- Zero hallucination

Auditable Provenance
- Table: AFL_Ontario_0
- Row: Row 0 (Tor. Eagles)
- Cell: City: Toronto
- Link: /wiki/Toronto
- Text: *"...town of York..."*

Reverse Grounding Fallback
- Maps answer back to graph coordinates
- Direct synthesis if disconnected
- Auditable evidence trail